# Ruoshui (Walter) Li

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## Education

**Saint Louis University** Fall 2025 - Present
M.A. in Mathematics, expected Spring 2027

**University of Connecticut** Sept. 2020 - Dec. 2024
B.A. in Mathematics

**East China Normal University** Sept. 2020 - Dec. 2021
Study abroad program during the COVID-19 pandemic

## Master's Thesis (in progress)

**Image Retrieval with DINO Features and SVD** (working title)

Supervisor: Vahan Huroyan | Expected completion: Spring 2027

- Ongoing study of whether dimensionality reduction and alternative representations of pretrained image features can improve image retrieval.
- Preliminary experiments compare whole-image, matrix, and subspace representations on Revisited Oxford and Paris using mean average precision. Project materials include saved comparison results and an interactive image-retrieval demonstration.

## Teaching Experience

**Precalculus Instructor (MATH 1400)** Spring & Fall 2026
Saint Louis University | Two semesters; Fall 2026 ongoing

- Introduce mathematical terminology and definitions, then use examples to help students understand the concepts.
- Prepare and refine lecture notes, worksheets, quizzes, examinations, and solutions; align assessments with taught material and classroom pacing.
- Use AI to assist with rubric-based exam grading and preparation of student feedback.

**Calculus I Teaching Assistant** Fall 2025
Saint Louis University

- Led discussion sessions, explained worked problems, and graded quizzes.

## Selected Academic Projects

**Multivariable Taylor's Theorem** 2026

- Class presentation for MATH 5023 on directional derivatives, multivariable Taylor's theorem, proof by reduction to one variable, and applications to error estimates and critical points.

**Calculating Normal Probabilities without Tables** Fall 2025

- Studied Bagby's 1995 approximation method using double integrals and polar coordinates.
- Gave a class presentation for MATH 5021 and a poster presentation in the Mentoring Math Minds (MMM) program at Saint Louis University.

## Skills

**Programming:** Python, R.

**AI-assisted problem solving:** Task decomposition, constraint setting, and iterative review.

- **Research:** Broke a retrieval study into nine key stages; specified resource constraints and candidate choices to guide AI-assisted testing of all planned combinations.
- **Grading:** Structured rubric-based workflows with sample review checkpoints, matched feedback files, and escalation of ambiguous cases.
- **Teaching materials:** Directed revisions to notes and assessments, checking content coverage, worked solutions, and classroom pacing.
- **Application planning:** Specified a workflow linking program requirements, shared questions, and school-specific essays, with review before submission.

**Additional background:** Completed four years at Sacred Heart High School, Waterbury, Connecticut; graduated in 2020.

Updated October 2026